



United International University (UIU)
Dept. of Computer Science and Engineering (CSE)
Mid Exam Year: 2021 Trimester: Fall
Course: CSE 2215/CSI 217 Data Structure and Algorithms I
Total Marks: 30, Time: 1 hour 45 min

(Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules)

There are FOUR questions. Answer all of them. Figures in the right-hand margin indicate full marks.

1. a) How does the descending order Quick Sort algorithm work on the following data? [4]

y p z x r

Here, x = last two digits of your student id + 1, $y = x + 3$, $z = x + y$, $p = y + z$, $r = x + 2$

- b) Discuss the time complexity of the following algorithm. [3]

```
sum=0;
for(i=1; i<=n; i++){
    for(j=1; j<=n; j++){
        sum=sum+i+j;
    }
}
printf("%d", sum);
```

2. a) Find the memory location of $A[15][20]$ if $\text{loc}(A[5][5]) = 8000 + c$, where c = last four digits of your student id. Assume row-wise memory is allocated in the integer array $A[50][60]$, where each integer data is 4 bytes. [2]

- b) How does the Binary Search algorithm work for the following data? Also find the total number of element comparisons needed in this case. [3]

Input Data: t r p z y x

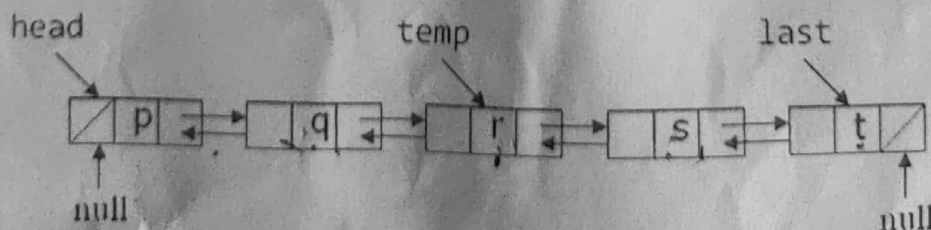
Search Key = y

Here, x = last two digits of your student id, $y = x + 3$, $z = x + y$, $p = y + z$, $r = z + p$, and $t = p + r$

- c) If $f(n) = kn^2 - 2$, prove that $f(n) = \Theta(n^2)$. Here, k = last digit of your student id + 3. [3]

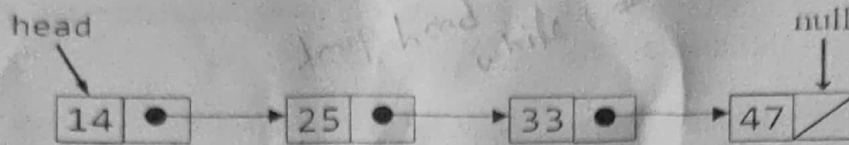
3. a) Answer the following questions for the doubly linked list as shown below, where p = last two digits of your student id + 5, $q = p + 3$, $r = p + q$, $s = r - 1$, $t = r + s$. [3]

- a) head $\rightarrow$ next $\rightarrow$ next $\rightarrow$ value = ?
b) last $\rightarrow$ prev $\rightarrow$ next $\rightarrow$ value = ?
c) temp $\rightarrow$ prev $\rightarrow$ prev $\rightarrow$ prev = ?
d) temp $\rightarrow$ next $\rightarrow$ prev $\rightarrow$ prev $\rightarrow$ value = ?
e) last $\rightarrow$ prev $\rightarrow$ prev $\rightarrow$ next $\rightarrow$ value = ?



b) Assume that you are given a single linked list as shown below. Write the statements to perform the following: [5]

- To insert 30 in between 25 and 33.
- To delete 47 from the list.
- To make a linear circular linked list from the current list.



4. a) Show the status of a STACK implemented by a linear linked list for the operations given below. Here, $x = \text{last digit of your student id} + 5$, $y = x + 3$, and $z = y + x$. [2]

Pop(), Push($x+y$), Push($y+z$), Pop(), Push($y*z$), Push($x*y$), Pop(), Pop()

- b) Show the status of a Queue implemented by an array of size, $m=2$ for the following operations. Here, Enqueue and Dequeue mean insert and delete respectively, and $x = \text{last digit of your student id} + 4$, $y = x + 1$, and $z = y + x$. [2]

Enqueue($x+y$), Enqueue($y+z$), Dequeue(), Enqueue($x+y$), Dequeue(), Dequeue()

- c) Show the manual tracing of the following algorithm using a Stack; [3]

```

#include<stdio.h>
#include<string.h>
int top=-1;
char Stack[6]={'\0'};
int main()
{
    char Str1[6]={'\0'};
    char Str2[6]={'\0'};
    int i;
    strcpy(Str1, "WORLD");
    for(i=0; i<5; ++i){
        Push(Str1[i]);
    }
    for(i=0; i<5; ++i){
        Str2[i]=Pop();
    }
    printf("%s", Str2);
    return 0;
}

void Push(char x){
    Stack[++top]=x;
    return;
}

char Pop(void){
    return Stack[top--];
}
  
```